1 Physics-aware encoding operator and fast gradients

We reconstruct images x on a grid $\Omega \subset \mathbb{R}^3$ from whitened multi-coil spiral data, without density compensation. The operator A contains off-resonance, R_2^*, fat, the concomitant field of the whole gradient history, gradient nonlinearity (GNL) and cardiac motion between readouts. It costs $L \approx 10$ FFT pairs per coil. Everything that is not a smooth function of the readout time is absorbed into a per-sample *chirp-aware stencil* (Section 1.3). Section 1.5 puts the velocity parametrisation on top of A, Section 1.6 gives the cost of one gradient $A^{\mathsf{H}}(Ax - y)$ and the levers that reduce it, and Section 1.7 gives the algebra of $\mathsf{N} = A^{\mathsf{H}}A$.

1.1 Signal model and phase split

With $\boldsymbol{r}$ the true scanner position of a voxel (after GNL, including the FOV offset), S_c the whitened coil maps, $\boldsymbol{k}_j$ the nominal trajectory and τ_j the time since excitation,

$$y_{cj} = \sum_{\boldsymbol{r}\in\Omega} S_c(\boldsymbol{r})\, x(\boldsymbol{r})\, \mathrm{e}^{-\mathrm{i}\Theta_j(\boldsymbol{r})} + n_{cj}, \qquad \Theta_j(\boldsymbol{r}) = \boldsymbol{k}_j\cdot\boldsymbol{r} + \big(\omega(\boldsymbol{r}) - \mathrm{i}R_2^*(\boldsymbol{r})\big)\tau_j + \sum_{q=1}^{4} w_{jq}\, m_q(\boldsymbol{r}), \quad (1)$$

with $n \sim \mathcal{CN}(0, I)$ after whitening by $\Psi^{-1/2}$. The concomitant monomials are $m = (z^2,\ x^2{+}y^2,\ xz,\ yz)$. A field with $\nabla\cdot\boldsymbol{B} = \nabla\times\boldsymbol{B} = 0$ has $\boldsymbol{B}_\perp = (G_x z - \frac{1}{2}G_z x,\ G_y z - \frac{1}{2}G_z y)$ and $|\boldsymbol{B}| \approx B_z + |\boldsymbol{B}_\perp|^2/2B_0$, so [2, 1]

$$\sum_q w_{jq} m_q(\boldsymbol{r}) = \frac{\gamma}{2B_0}\big\langle W_j,\ M_{\boldsymbol{r}}^\top M_{\boldsymbol{r}}\big\rangle, \qquad W_j = \int_0^{t_j} \boldsymbol{G}\boldsymbol{G}^\top \mathrm{d}t, \qquad M_{\boldsymbol{r}} = \begin{pmatrix} z & 0 & -x/2 \\ 0 & z & -y/2 \end{pmatrix}, \quad (2)$$

with relative error $O(|\boldsymbol{B}_\perp|^2/B_0^2) \approx 2\cdot 10^{-4}$ at 3 T. Here $\boldsymbol{G}$ is the physical gradient (after rotation and FOV offset) and W_j integrates from excitation, so the bipolars, the k_z blip and the readout all contribute.

Split. Let $\bar{w}_q(\tau)$ be the mean of w_{jq} over the readouts of an encode at ADC time τ, and $\Delta w_{jq} = w_{jq} - \bar{w}_q(\tau_j)$. Then

$$\Theta_j(\boldsymbol{r}) = \boldsymbol{k}_j\cdot\boldsymbol{r} + \psi(\boldsymbol{r}, \tau_j) + \Delta_j(\boldsymbol{r}), \qquad \psi(\boldsymbol{r}, \tau) = \big(\omega - \mathrm{i}R_2^*\big)\tau + \sum_q \bar{w}_q(\tau) m_q, \qquad \Delta_j = \sum_q \Delta w_{jq} m_q. \quad (3)$$

ψ is an arbitrary spatial function of $(\boldsymbol{r}, \tau)$ and is carried exactly by time segments (Section 1.2). Δ_j is small and polynomial. In logical coordinates $\boldsymbol{\rho}$ (GNL-warped positions $\boldsymbol{r}(\boldsymbol{\rho})$) we fit each $m_q(\boldsymbol{r}(\boldsymbol{\rho}))$ over the support with monomials of degree ≤ 4:

$$\Delta_j(\boldsymbol{\rho}) = \phi_{0j} + \boldsymbol{b}_j\cdot\boldsymbol{\rho} + \boldsymbol{\rho}^\top \Delta C_j\, \boldsymbol{\rho} + c_j(\boldsymbol{\rho}) + \varepsilon_j(\boldsymbol{\rho}), \qquad c_j(\boldsymbol{\rho}) = \sum_q \Delta w_{jq} \sum_{3\le|\alpha|\le 4} h_{q\alpha}\boldsymbol{\rho}^\alpha. \quad (4)$$

The constant ϕ_{0j} becomes a sample phase and the linear part an exact shift $\boldsymbol{k}'_j = \boldsymbol{k}_j + \boldsymbol{b}_j$. The chirp $\chi_j = \boldsymbol{\rho}^\top \Delta C_j \boldsymbol{\rho} + c_j$ goes into the stencil. Phase budget of the full concomitant field, simulated 3 T spiral (80 mT/m, 36 cm FOV, 4.4 ms readout, 3 cm off-centre) [measured]:

	constant ϕ_0	linear $\boldsymbol{b}\cdot\boldsymbol{\rho}$	quadratic (corners)
max over samples and FOV	0.08 rad	0.96 rad (≤ 0.29 cells)	3.1 rad

Without GNL the quadratic fit residual ε_j is zero. At GNL-warped positions it is $4.5\cdot 10^{-3}$ rad rms (0.056 max) for degree 2, and degree 4 removes most of it (table in Section 1.3).

1.2 Time segments

With segment diagonals $D_\ell(\boldsymbol{r}) = \mathrm{e}^{-\mathrm{i}\psi(\boldsymbol{r},\tau_\ell)}$, $\ell = 1..L$, the interpolators are the least-squares solution over the support Ω_s [3]:

$$\begin{aligned} \boldsymbol{a}(\tau) = \operatorname*{arg\,min}_{\boldsymbol{a}\in\mathbb{C}^L} \sum_{\boldsymbol{r}\in\Omega_s} \left| \mathrm{e}^{-\mathrm{i}\psi(\boldsymbol{r},\tau)} - \sum_\ell a_\ell D_\ell(\boldsymbol{r}) \right|^2 \iff G\boldsymbol{a}(\tau) = \boldsymbol{\beta}(\tau), \\ G_{\ell\ell'} = \langle D_\ell, D_{\ell'} \rangle_{\Omega_s}, \qquad \beta_\ell(\tau) = \langle D_\ell, \mathrm{e}^{-\mathrm{i}\psi(\cdot,\tau)} \rangle_{\Omega_s}. \end{aligned} \tag{5}$$

$\boldsymbol{a}$ is tabulated at every ADC sample, which removes a 3–7 $\cdot 10^{-4}$ table floor. The rank is set by the number of phase cycles $\Delta f\, T_{\mathrm{ro}}$ (Δf the 1–99 % off-resonance span), not by resolution. Default: $L = \lceil 1.3\, \Delta f\, T_{\mathrm{ro}} + 3.5 \rceil$. Model error against an f64 direct sum, smooth 450 Hz field [measured]:

L	2 ms	4 ms	8 ms	12 ms
4	$9 \cdot 10^{-4}$			
6	$8 \cdot 10^{-6}$	$2.5 \cdot 10^{-4}$	$1.5 \cdot 10^{-2}$	
8			$6 \cdot 10^{-4}$	
10			$2.7 \cdot 10^{-5}$	$8.9 \cdot 10^{-4}$

Species and decay in the diagonals. Complex ψ carries R_2^* at no cost. Water and fat with a known fat fraction $f(\boldsymbol{r})$ and the six-peak spectrum $c(\tau)$ share one set of segments:

$$D_\ell(\boldsymbol{r}) = g_{\boldsymbol{r}}(\tau_\ell), \qquad g_{\boldsymbol{r}}(\tau) = \mathrm{e}^{-\mathrm{i}\psi(\boldsymbol{r},\tau)} \big(1 - f(\boldsymbol{r}) + f(\boldsymbol{r})\, c(\tau)\big). \tag{6}$$

With this, $L = 12$ gives $5.0 \cdot 10^{-4}$ and $L = 14$ gives $6.3 \cdot 10^{-5}$. Two species with separate segments ($2L = 16$ paths) give $6.5 \cdot 10^{-4}$ [measured, CPU].

1.3 Chirp-aware stencils

After Equations (3) and (4), with $v_{c\ell} = S_c D_\ell x$, sample j needs the *modulated row*

$$(Ax)_{cj} = \mathrm{e}^{-\mathrm{i}\phi_{0j}} \sum_\ell a_\ell(\tau_j) \sum_{\boldsymbol{r}} v_{c\ell}(\boldsymbol{r})\, \mathrm{e}^{-\mathrm{i}\boldsymbol{k}'_j\cdot\boldsymbol{r}}\, \mathrm{e}^{-\mathrm{i}\chi_j(\boldsymbol{r})}. \tag{7}$$

The NUFFT represents a plane wave on Ω through the oversampled grid $\boldsymbol{u} \in \mathbb{Z}^3/(\sigma N)$ and a window φ with apodisation $\hat{\varphi}$ [5, 6]: $\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} = \hat{\varphi}(\boldsymbol{r})^{-1} \sum_{\boldsymbol{u}} \varphi(\boldsymbol{k} - \boldsymbol{u}) \mathrm{e}^{-\mathrm{i}\boldsymbol{u}\cdot\boldsymbol{r}} + \epsilon$. A modulated row needs a different set of taps:

Definition 1 (chirp-aware stencil)**.** $\Gamma_j : \mathbb{Z}^3 \to \mathbb{C}$*, supported on* $\|\boldsymbol{u} - \boldsymbol{k}'_j\|_\infty \le J/2$*, with*

$$\sum_{\boldsymbol{u}} \Gamma_j(\boldsymbol{u})\, \mathrm{e}^{-\mathrm{i}\boldsymbol{u}\cdot\boldsymbol{r}} \;\approx\; \hat{\varphi}(\boldsymbol{r})\, \mathrm{e}^{-\mathrm{i}\boldsymbol{k}'_j\cdot\boldsymbol{r}}\, \mathrm{e}^{-\mathrm{i}\chi_j(\boldsymbol{r})}, \qquad \boldsymbol{r} \in \Omega. \tag{8}$$

Then, with $G_{c\ell} = \mathcal{F}_\sigma[\hat{\varphi}^{-1} S_c D_\ell x]$ (one oversampled FFT per channel),

$$(Ax)_{cj} = \mathrm{e}^{-\mathrm{i}\phi_{0j}} \sum_{\boldsymbol{u}} \Gamma_j(\boldsymbol{u}) \sum_\ell a_\ell(\tau_j)\, G_{c\ell}(\boldsymbol{u}), \qquad \big(A^{\mathsf{H}} y\big) = \sum_c \overline{S_c} \sum_\ell \overline{D_\ell}\, \hat{\varphi}^{-1} \mathcal{F}_\sigma^{\mathsf{H}} \Big[\sum_j \overline{\Gamma_j}\, \overline{a_\ell(\tau_j)}\, \mathrm{e}^{\mathrm{i}\phi_{0j}} y_{cj} \Big]. \tag{9}$$

Γ_j *depends on neither the coil nor the segment.* It is evaluated once per (sample, tap) and reused for all $C \cdot L$ channels. A^{H} is the exact transpose. A plain NUFFT has $\Gamma_j = \varphi(\boldsymbol{k}'_j - \cdot)$. We use two constructions of Γ_j.

(a) Taylor stencils: derivative weights of the window (production)

From $\rho^\alpha e^{-i\boldsymbol{k}\cdot\boldsymbol{\rho}} = (i\partial_{\boldsymbol{k}})^\alpha e^{-i\boldsymbol{k}\cdot\boldsymbol{\rho}}$, any polynomial phase acts on the plane wave as a differential operator in $\boldsymbol{k}$:

$$e^{-i\chi(\boldsymbol{\rho})}e^{-i\boldsymbol{k}\cdot\boldsymbol{\rho}} = \sum_{n=0}^{P} \frac{(-i)^n}{n!}\,\chi(i\partial_{\boldsymbol{k}})^n\, e^{-i\boldsymbol{k}\cdot\boldsymbol{\rho}} + R_P, \qquad |R_P| \le \frac{|\chi|^{P+1}}{(P+1)!}. \tag{10}$$

Applied to the interpolant this gives the stencil in closed form:

Proposition 2 (Taylor stencil)**.** *Let* $\Delta C : \nabla\nabla = \sum_{pq} \Delta C_{pq}\partial_p\partial_q$*. Then*

$$\Gamma_j^{(P)}(\boldsymbol{u}) = \Big[\,1 + i\,\Delta C_j : \nabla\nabla - \tfrac{1}{2}\big(\Delta C_j : \nabla\nabla\big)^2 - i\sum_{3\le|\alpha|\le 4} i^{|\alpha|}\Big(\sum_q \Delta w_{jq} h_{q\alpha}\Big)\partial^\alpha\Big]\varphi(\boldsymbol{\kappa}-\boldsymbol{u})\Big|_{\boldsymbol{\kappa}=\boldsymbol{k}'_j} \tag{11}$$

($P = 2$ uses the first three terms, and order 3 adds the last term to first order). The result is the exact derivative of the interpolant $\sum_{\boldsymbol{u}} \varphi(\boldsymbol{k}-\boldsymbol{u})G(\boldsymbol{u})$, not a finite difference. The error is $|\chi_j|^{P+1}/(P+1)!$ plus the window error of $\partial^\alpha\varphi$.

Proof. $\chi = \boldsymbol{\rho}^\top \Delta C \boldsymbol{\rho} \Rightarrow \chi(i\partial) = -\Delta C : \nabla\nabla$. Insert into Equation (10) for $n \le 2$, use $\rho^\alpha \mapsto i^{|\alpha|}\partial^\alpha$ for c_j, and move $\partial_{\boldsymbol{k}}$ onto $\varphi(\boldsymbol{k}-\boldsymbol{u})$. (Checked against the exact chirped sum.) □

Separable evaluation. $\partial^\alpha\varphi(\boldsymbol{\kappa}-\boldsymbol{u}) = \prod_a \varphi^{(\alpha_a)}(\kappa_a - u_a)$, so

$$\Gamma_j^{(P)}(\boldsymbol{u}) = \sum_{|\alpha|\le 4} \gamma_{j\alpha} \prod_{a=1}^{3} \varphi^{(\alpha_a)}(\kappa_{ja} - u_a), \qquad \gamma_j \in \mathbb{C}^{35}. \tag{12}$$

Each tap is a sum of 35 rank-one products of 1-D tables $\varphi^{(0..4)}$ (5 values per axis per tap). The coefficients come from the 6 numbers of ΔC_j, their 15 distinct fourth-order products, and $u_j \in \mathbb{R}^4$ with a 4×25 table h. Everything is formed in registers. *Window.* Order 2 needs $\varphi^{(4)}$ bounded. The exponential of semicircle $e^{\beta(\sqrt{1-z^2}-1)}$ has $\sqrt{\cdot}$-singular derivatives at $|z| = 1$, so they are clamped to 0 for $|z| > 0.98$, where $\varphi < 10^{-7}$; without the clamp the fourth-derivative error is $46\times$ larger. Interpolated value and ∂^2 errors (1-D) [measured]:

σ (grid volume vs. σ=2)	W	value	∂^2
2.0 (1.00)	7	$9\cdot10^{-7}$	$8\cdot10^{-5}$
1.5 (0.42)	10	$6\cdot10^{-8}$	$6\cdot10^{-5}$
1.25 (0.24)	10	$2\cdot10^{-6}$	$3\cdot10^{-5}$

The correction term is $\lesssim 0.2$ rad, so ∂^2 accuracy 10^{-3} suffices: $\sigma = 1.25$, $W = 10$. *Accuracy* vs. an f64 direct sum (B0 + τ-dependent quadratic + per-sample $\Delta C + \phi_0$) [measured]:

stencil order P	$L = 6$	$L = 8$	$L = 10$
0 (plain window)	$3.8\cdot10^{-2}$	–	–
1	$2.4\cdot10^{-3}$	$1.7\cdot10^{-3}$	$1.7\cdot10^{-3}$
2	$1.8\cdot10^{-3}$	$8.5\cdot10^{-5}$	$7.7\cdot10^{-5}$

Each order has a floor set by the dropped next term, independent of L. Full model with GNL ($L = 20$, forward / adjoint) [measured, CPU emulation]:

B_0	2.89 T	1.5 T	0.55 T
order 2 (degree-2 fit)	$4.8 \cdot 10^{-3}$ / $8.3 \cdot 10^{-3}$	$8.9 \cdot 10^{-3}$ / $1.5 \cdot 10^{-2}$	$3.1 \cdot 10^{-2}$ / $7.0 \cdot 10^{-2}$
order 3 (degree-4 fit)	$1.2 \cdot 10^{-3}$ / $2.2 \cdot 10^{-3}$	$3.0 \cdot 10^{-3}$ / $6.5 \cdot 10^{-3}$	$2.9 \cdot 10^{-2}$ / $7.3 \cdot 10^{-2}$

At 0.55 T, $|\chi_j|$ reaches 1–1.4 rad, which is beyond the Taylor range ($|\chi|^3/6$). That is the regime of construction (b).

(b) Least-squares stencils for large chirps

For $|\chi| \gtrsim 1$ we fit Equation (8) directly on a lattice Ω_h of the FOV, with $\Phi_{\boldsymbol{w}} = \sum_q w_q m_q$:

$$\Gamma(\cdot; \boldsymbol{w}, \boldsymbol{\kappa}) = \underset{\operatorname{supp}\Gamma \subset \boldsymbol{\kappa} + [-S,S]^3}{\arg\min} \sum_{\boldsymbol{r} \in \Omega_h} \left| \sum_{\boldsymbol{u}} \Gamma(\boldsymbol{u}) \mathrm{e}^{-\mathrm{i}\boldsymbol{u}\cdot\boldsymbol{r}} - \hat{\varphi}(\boldsymbol{r}) \mathrm{e}^{-\mathrm{i}\boldsymbol{\kappa}\cdot\boldsymbol{r} - \mathrm{i}\Phi_{\boldsymbol{w}}(\boldsymbol{r})} \right|^2 + \varrho \|\Gamma\|^2. \tag{13}$$

Write $\Gamma = \varphi \star \hat{g}(\cdot; \boldsymbol{w})$, where $\hat{g}$ is the compact spectrum of the chirp over the FOV. The family $\{\hat{g}(\cdot; \boldsymbol{w})\}_{\boldsymbol{w} \in \mathcal{W}}$ is compressed by a POD basis $\{B_l\}_{l \le L_p}$ with DEIM points $\{\boldsymbol{r}_i\}$:

$$\hat{g}(\cdot; \boldsymbol{w}) \approx \sum_{l=1}^{L_p} \beta_l(\boldsymbol{w}) B_l, \qquad \boldsymbol{\beta}(\boldsymbol{w}) = \left[B_l(\boldsymbol{r}_i)\right]_{il}^{-1} \left[\mathrm{e}^{-\mathrm{i}\Phi_{\boldsymbol{w}}(\boldsymbol{r}_i)}\right]_i. \tag{14}$$

Per sample this costs L_p sincos, one $L_p \times L_p$ solve and an $L_p \times (2S{+}1)^3$ contraction, followed by a separable convolution with φ. The fitted stencil is much narrower than a DFT of a windowed chirp: $S = 3$–4 instead of 8–10 for 10^{-3} at 1 rad. Training on the sequence's own $\boldsymbol{w}_j$ instead of the coefficient box gives $L_p = 31 \to 13$. Taps per axis J at tolerance 10^{-3} (1-D, least-squares stencil with a shared correction function) [measured]:

$\Phi_{\max}$ [rad]	0	1	2	4	8
J at $\sigma = 2$	4	8	10	13	18
J at $\sigma = 1.25$	8	8	10	14	18

At $\sigma = 1.25$ a chirp up to about 1 rad is free, because the plain stencil is already 8 wide. Beyond that $J \approx \max(J_{\text{plain}}, \text{chirp spread})$. In k-space the Gram matrix is banded, $\hat{\mathsf{N}} = \Gamma^{\mathsf{H}}\Gamma$, with bandwidth equal to the stencil support. The diagonal fields (z^2, $x^2{+}y^2$) give an exactly separable $\hat{g}$; with the cross terms the CP rank is 3–4 at 10^{-3}. *Status* [measured, $N = 128$, $K = 10^6$, $C = 8$, $S = 3$]: $A^{\mathsf{H}}A = 18.2$ ms/coil (39 plain-Toeplitz applications) with stored fp16 stencils and a tiled scatter. This is the only construction with one FFT pair per coil independent of Φ and L. It handles the concomitant phase only, with no B0 segments yet.

Choice

construction	FFT pairs/coil	taps/sample	valid for	status
Taylor, $P = 2$	L	W^3	$\lvert\chi\rvert \lesssim 0.7$	production
Taylor, order 3	L	W^3	+ GNL residual	built
least squares	1	$(2S{+}W)^3$	$\Phi \lesssim 8$	built, no B0
Weyl/Toeplitz (Section 1.7)	4–35	0	$\Phi \lesssim 3$	built, unused

Not built: least-squares stencils for the post-split residual χ_j inside the segmented operator. That would combine L B0 segments with $|\chi| > 1$ (low field, strong GNL).

1.4 The full operator

Time-resolved sets add a periodic B-spline in cardiac time, $x(\boldsymbol{r},t)=\sum_f B_f(t)x_f(\boldsymbol{r})$, linear ($N_f=2$) or cubic ($N_f=4$):

$$(Ax)_{cj}=\mathrm{e}^{-\mathrm{i}\phi_{0j}}\sum_{\boldsymbol{u}}\Gamma_j(\boldsymbol{u})\sum_{f\in\mathcal{F}(t_j)}B_f(t_j)\sum_{\ell=1}^{L}a_\ell(\tau_j)\,G_{cf\ell}(\boldsymbol{u}),\qquad G_{cf\ell}=\mathcal{F}_\sigma\big[\hat{\varphi}^{-1}S_cD_\ell x_f\big]. \tag{15}$$

FFTs scale with the number of frames and gather/scatter with N_f. Several images on the same samples share one tile pass: $y=\sum_k w_k\odot Ax_k$, $x_k=A^{\mathsf{H}}(\bar{w}_k\odot y)$, $k\le 4$ (separate-species fat: $w_{\text{fat}}=c(\tau_j)$). The adjoint accumulates per tile without atomics (exact CSR tile→sample lists). Cost per call, RTX 4090, $180\times180\times28$, $M=2.2\cdot10^5$, 12 coils [measured, ms]:

operator	forward	adjoint
plain NUFFT, σ=2, W=6	6.0	24.7
segmented, L=1, P=0	5.7	8.7
segmented, L=8, $P=0/1/2$	30.1 / 30.9 / 31.7	31.2 / 31.8 / 30.1
segmented, L=10/12, P=2	40.4 / 49.2	35.3 / 41.0

The $P=2$ stencil costs the same as $P=0$ because Γ_j is shared by all $C\cdot L$ channels. About half of the call is the memory-bound pad/FFT/crop.

1.5 Velocity parametrisation

Model. With encode moments $\boldsymbol{k}_e=\gamma\boldsymbol{M}_{1,e}$ and temporal bases $\beta^z_\kappa,\beta^v_\kappa$ (periodic B-splines, independent of the operator frames),

$$x_e(\boldsymbol{r},t)=z(\boldsymbol{r},t)\,\mathrm{e}^{\mathrm{i}\varphi_e(\boldsymbol{r})}\,\mathrm{e}^{-\mathrm{i}\boldsymbol{k}_e\cdot\boldsymbol{v}(\boldsymbol{r},t)},\quad z=\sum_{\kappa\le K_z}\beta^z_\kappa(t)z_\kappa,\quad \boldsymbol{v}=\sum_{\kappa\le K_v}\beta^v_\kappa(t)\boldsymbol{v}_\kappa,\quad \varphi_e=\sum_p a_{ep}P_p. \tag{16}$$

Here $z\in\mathbb{C}$ carries the magnitude and every phase common to all encodes, $\boldsymbol{v}\in\mathbb{R}^3$ is the velocity, and P_p are Legendre polynomials (residual eddy-current or model phase, degree ≤ 3). The unknowns $\theta=(z_\kappa,\boldsymbol{v}_\kappa,a)$ are 5 reals per voxel and node instead of $2E$. The model is linear in z, and dual-VENC unwrapping is part of the solve. The operator sees images at frames t_f: $x_{ef}=x_e(\cdot,t_f)$. Interpolation to each sample's t_j happens inside A (Equation (15)). Each (frame, encode) set $s=(f,e)$ has its own A_s, and

$$F(\theta)=\tfrac{1}{2}\sum_s\omega_s\big\|A_s\,x_s(\theta)-y_s\big\|^2+\tfrac{\lambda}{2}\sum_\kappa\big\|w^{1/2}\boldsymbol{v}_\kappa\big\|^2, \tag{17}$$

where ω_s are the VENC-continuation weights and w a static-tissue mask (robust IRLS fit of the warm-start phase differences, refined by pulsatility).

Gradient. One gradient is one image-space gradient per set followed by a pointwise vector–Jacobian product. With $g_s=A_s^{\mathsf{H}}(A_sx_s-y_s)$ and $u_s=\mathrm{e}^{\mathrm{i}\varphi_e-\mathrm{i}\boldsymbol{k}_e\cdot\boldsymbol{v}(t_f)}$ (so $x_s=z(t_f)u_s$),

$$\begin{aligned}\nabla_{z_\kappa}F&=\sum_s\omega_s\beta^z_\kappa(t_f)\,\overline{u_s}\,g_s,\qquad \nabla_{a_{ep}}F=-\sum_{s\ni e}\omega_s\sum_{\boldsymbol{r}}P_p\,\mathrm{Im}\big(\overline{g_s}\,x_s\big),\\ \nabla_{\boldsymbol{v}_\kappa}F&=\sum_s\omega_s\beta^v_\kappa(t_f)\,\boldsymbol{k}_e\,\mathrm{Im}\big(\overline{g_s}\,x_s\big)+\lambda w\boldsymbol{v}_\kappa.\end{aligned} \tag{18}$$

Proof: $\mathrm{d}x_s = u_s\,\mathrm{d}z - \mathrm{i}(\boldsymbol{k}_e\cdot\mathrm{d}\boldsymbol{v})x_s + \mathrm{i}P_p x_s\,\mathrm{d}a_{ep}$ and $\mathrm{d}F = \mathrm{Re}\langle g_s, \mathrm{d}x_s\rangle$. The cost is that of the complex model (one forward and one adjoint per set) plus $O(|\Omega|K)$ pointwise work. It does not grow with K_z, K_v.

Metric. Gauss–Newton $H = J^{\mathsf{H}}\mathsf{N}J$ with $\mathsf{N} \approx \hat{n}I$, where $\hat{n} = \mathbb{E}_\xi\langle\xi, \mathsf{N}\xi\rangle/\|\xi\|^2$ per unit coil energy (random probe), gives pointwise blocks per node:

$$M_z^{-1} = \hat{n}\sum_s \omega_s \beta_\kappa^z(t_f)^2, \qquad M_{\boldsymbol{v}}^{-1}(\boldsymbol{r}) = \hat{n}\,|\bar{z}(\boldsymbol{r})|^2 \sum_s \omega_s \beta_\kappa^v(t_f)^2\,\boldsymbol{k}_e\boldsymbol{k}_e^\top + \lambda w(\boldsymbol{r})I + \mu I. \tag{19}$$

These are refreshed every 5 gradients and used as the base metric of ZMSHR1 (Section 1.8). The cross term $\partial^2 F/\partial z\partial\boldsymbol{v} \propto |z|^2 \sin\phi_c \sum_e \boldsymbol{k}_e$ (with ϕ_c an encode-common phase error) is dropped by the block metric. For referenced and dual-VENC encodings $\sum_e \boldsymbol{k}_e \neq 0$, so the solve starts with n z-only iterations, which form a linear subproblem.

Exact phases on the sample side. A degree-1 residual phase is a k-space shift: $A_{\boldsymbol{k}}(x\,\mathrm{e}^{\mathrm{i}(c+\boldsymbol{g}\cdot\boldsymbol{\rho})}) = \mathrm{e}^{\mathrm{i}c}A_{\boldsymbol{k}-\boldsymbol{g}}(x)$, verified to $5\cdot10^{-7}$. Injected or estimated linear phases therefore cost nothing. GNL: reconstruct at apparent positions $\boldsymbol{\rho} = u(\boldsymbol{r})$ and correct $\boldsymbol{v}_{\text{true}} = J_u^{-1}\boldsymbol{v}_{\text{app}}$.

Readout motion (not yet in the reconstruction). During the readout the first moment adds $\mathrm{e}^{-\mathrm{i}\gamma\boldsymbol{m}_{1,j}(\tau)\cdot\boldsymbol{v}(\boldsymbol{r})}$. The simulator includes it, so it is a model mismatch of the current reconstruction. It has the form of $\omega(\boldsymbol{r})\tau$. With $\boldsymbol{m}_{1,j} = \bar{\boldsymbol{m}}_1(\tau) + \delta\boldsymbol{m}_{1,j}(\tau)$ (mean over interleaves plus a rotation-dependent part), the planned treatment is

$$\psi(\boldsymbol{r},\tau) \;\leftarrow\; \psi(\boldsymbol{r},\tau) + \gamma\,\bar{\boldsymbol{m}}_1(\tau)\cdot\boldsymbol{v}^{(k)}(\boldsymbol{r}), \tag{20}$$

which is lagged in the iterate $\boldsymbol{v}^{(k)}$, uses the same segments, and refreshes $D_\ell, \boldsymbol{a}$ when $\|\boldsymbol{v} - \boldsymbol{v}^{(k)}\|$ grows. The rotation-dependent part $\delta\boldsymbol{m}_{1,j}\cdot\boldsymbol{v}(\boldsymbol{r})$ is not polynomial in $\boldsymbol{r}$ and is not modelled.

Results (R4, 7-point dual-VENC 1.0/3.5 m/s, 60 s, $R \approx 8$ per frame, 30 iterations, no prior) [measured]:

model	velocity RMSE (m/s)	wrapped
complex images + locally-low-rank prior	0.183	0.4 %
Equation (16), $K_z = K_v = 8$ cubic, VENC continuation	0.138	0.11 %
+ grid 1.25× finer (1.6 mm)	0.130	–
+ injected residual phase 0.6 rad, robust init + anchor	0.166	–
4-point, full physics + GNL, 10 z-only + 30 joint	0.232	0.01 %

1.6 Evaluating the gradient $A^{\mathsf{H}}(Ax - y)$ fast

Cost model. The gradient is $\nabla f = \sum_s A_s^{\mathsf{H}}(A_s x_s - y_s)$, computed as forward, residual, adjoint. With $|\Omega_\sigma| = \sigma^3|\Omega|$ grid points and M_s samples,

$$\begin{aligned} T_\nabla \approx \sum_s \Big[&\underbrace{2CF_sL\left(T_{\text{FFT}}(|\Omega_\sigma|) + T_{\text{pad/crop}}\right)}_{\text{grid side} \propto CF_sL|\Omega_\sigma|} + \underbrace{2CL\,N_f M_s W^3 t_{\text{mac}}}_{\text{gather/scatter}} \\ &+ \underbrace{N_f M_s W^3\, n_\gamma t_{\text{mac}}}_{\text{stencil, coil- and segment-free}} + \underbrace{3CM_s\,t_{\text{mem}}}_{\text{residual}} \Big]. \end{aligned} \tag{21}$$

Here F_s is the number of frame grids that set s touches (1 for a static set), and $n_\gamma \leq 35$.

Implemented levers.

1. *Shared stencils.* The n_γ term is divided over CLN_f channels, so $P = 2$ costs the same as $P = 0$ (31.7 vs. 30.1 ms).

2. *Small grid.* Bounded window derivatives allow $\sigma = 1.25$: $|\Omega_\sigma| = 0.24\times$ the $\sigma = 2$ grid.

3. *Fused grid passes.* The grid is channel-interleaved [point][$c \cdot L + \ell$] with one partial-axis FFT over all channels. The pad pass forms $\hat{\varphi}^{-1} S_c D_\ell x_f$ for all (c, ℓ) in one pass; the crop pass forms $\sum_c \overline{S_c} \sum_\ell \overline{D_\ell}(\cdot)$ in one multi-input pass. Launch shapes are planned within 48 KB of shared memory and autotuned per set.

4. *Coil compression.* With the whitened Gram $\sum_n y_n y_n^{\mathsf{H}} = U\Sigma^2 U^{\mathsf{H}}$, keep $C_v = \min\{n : \sum_{i\le n} s_i^2 \ge 0.999 \sum_i s_i^2\}$ virtual coils $U_{:,1:C_v}^{\mathsf{H}} y$. Rows are orthonormal, so the noise stays white. On the 32-coil array: $C_v = 16$ and 29 [measured]. The Gram is taken over the Cartesian calibration region: a centre-dense trajectory would weight it towards $k = 0$ and keep too few coils.

5. *Stochastic coil subsets.* With energy-ordered virtual coils, keep the leading set $\mathcal{K}$ and draw $\mathcal{D}_k$ from the remaining ones:

$$\tilde{\nabla} f = \sum_{c\in\mathcal{K}} g_c + p \sum_{c\in\mathcal{D}_k} g_c, \qquad \mathbb{E}\,\tilde{\nabla} f = \nabla f, \tag{22}$$

   where g_c is the gradient of coil c and p is the inverse inclusion probability. With random reshuffling the epoch mean is exact. The step uses $\alpha = 1/\max\big(\lambda_{\max}(\mathsf{N}), \max_{\text{draws}} \lambda_{\max}(\mathsf{N}_\mathcal{K} + p\mathsf{N}_\mathcal{D})\big)$. A subset is held for $h \ge 2$ iterations, so the ZMSHR1 secant pair satisfies $u = \tilde{\mathsf{N}} s$ exactly. Measured on 8 virtual coils: the epoch mean equals the full gradient to $7 \cdot 10^{-6}$; $\lambda_{\max}$ inflation is 1.42 when drawing 4 of 8 and 0.90 when keeping 2 and drawing 3 of 6. At 60 gradient-equivalents PnP-ISTA has 10–16 % lower error; ZMSHR1 does not gain.

6. *Shared gridding* of $k \le 4$ images on the same samples (Section 1.4). One-set water+fat segments use L channels instead of $2L$ (Section 1.2).

7. *Streaming.* The k-space data y_s come from a host store (dense, lz4/zstd blocks, or file). Sample tables are kept packed (zstd, $\times 6.2$) and decoded into pinned staging buffers. A worker decodes set $s{+}1$ while the GPU runs set s, so the wall time per set is $\max(T_{\text{GPU}}, T_{\text{decode}})$ rather than their sum. A coil subset reads only its own rows.

8. *Lagged control variate with a reduced-segment surrogate.* The surrogate $\tilde{E}$ is the same operator with $L_\text{s} < L$ segments on the same sample tables, and $\tilde{\mathsf{N}} = \tilde{E}^{\mathsf{H}}\tilde{E}$:

$$\begin{aligned} \tilde{g}_k &= \tilde{\mathsf{N}}(x_k - x_a) + \nabla f(x_a), \qquad x_a \leftarrow x_k \text{ if } \|x_k - x_a\| > \tau\|x_k\|, \\ \tilde{f}_k &= f(x_a) + \mathrm{Re}\langle \nabla f(x_a), x_k - x_a\rangle + \tfrac{1}{2}\|\tilde{E}(x_k - x_a)\|^2. \end{aligned} \tag{23}$$

   The error is $(\tilde{\mathsf{N}} - \mathsf{N})(x_k - x_a)$, which vanishes at the fixed point. Cheap steps read no data and cost $\approx L_\text{s}/L$ of an exact step. A secant pair is kept only within one anchor (or between two exact gradients); otherwise the metric step is used. L_s is chosen as the smallest value with $\|(E - \tilde{E})x\| \le 0.03\|Ex\|$ on the warm start. Measured (test operator, 60 ZMSHR1 iterations, $\tau = 0.1$): with a mismatch of 2.4 % (1.5 phase cycles, $L_\text{s} = 3$ of 8) or 15 % at 0.3 cycles, 7 exact + 53 cheap gradients reach the exact run's error. With 7–8 % at ≥ 1.5 cycles it needs 33–45 exact gradients and ends 2–3$\times$ further away. On sim_point (4.3 cycles, $L = 10$, chosen $L_\text{s} = 4$ at 2.6 %), 30 iterations took 10 exact + 20 cheap gradients and 28 % less time, at a 35 % higher final f.

Not yet implemented.

1. *Fused residual.* The gather accumulates $y \mathrel{+}= \Gamma G$, so initialising the accumulator with $-y_s$ instead of 0 yields $r_s = A_s x_s - y_s$ directly. A reduction in the same kernel gives $\frac{1}{2}\|r_s\|^2$. This removes the $3CM_s$ term and the residual buffer, or keeps only one coil batch of it.

2. *Pruned FFT.* $\hat{\varphi}^{-1} S_c D_\ell x$ fills N of σN lines per axis. Pruning the first two passes of a three-pass FFT leaves the fraction $\frac{1}{3}(\sigma^{-2} + \sigma^{-1} + 1)$: 0.81 at $\sigma = 1.25$ and 0.58 at $\sigma = 2$.

3. *Segment Gram (Toeplitz) form* [4]:

$$\mathsf{N} = \sum_c \overline{S_c} \sum_{\ell\ell'} \overline{D_\ell}\, \mathsf{T}[g_{\ell\ell'}]\, D_{\ell'} S_c, \qquad g_{\ell\ell'}(\boldsymbol{\Delta}) = \sum_j \overline{a_\ell(\tau_j)} a_{\ell'}(\tau_j) \mathrm{e}^{\mathrm{i}\boldsymbol{k}_j \cdot \boldsymbol{\Delta}} \quad \text{(no per-sample chirp).}$$

   It needs $L(L+1)/2$ kernels of $(2N)^3$ per set, which does not fit for time-resolved sets (≈ 80 sets). It is useful for few-set problems.

1.7 Algebra of the normal operator

Coil maps are pointwise multiplications, so $\mathsf{N} = \sum_c \overline{S_c}\, \mathsf{N}_0\, S_c$ with N_0 coil-free. Below, $\mathsf{N} := \mathsf{N}_0$, $\Phi_j = \Theta_j - \boldsymbol{k}_j \cdot \boldsymbol{r}$, and X_a, $\mathsf{P} = -\mathrm{i}\nabla$, $\mathsf{T}[g]\rho = g * \rho$:

$$\mathsf{N}(\boldsymbol{r}, \boldsymbol{r}') = \sum_j \mathrm{e}^{\mathrm{i}\boldsymbol{k}_j \cdot (\boldsymbol{r} - \boldsymbol{r}')} \mathrm{e}^{\mathrm{i}(\Phi_j(\boldsymbol{r}) - \Phi_j(\boldsymbol{r}'))}, \qquad \mathsf{N} = \sum_j \mathsf{Q}_j\, \mathsf{T}[\mathrm{e}^{\mathrm{i}\boldsymbol{k}_j \cdot \boldsymbol{\Delta}}]\, \mathsf{Q}_j^{\mathsf{H}}, \quad \mathsf{Q}_j = \mathrm{e}^{\mathrm{i}\Phi_j(\mathsf{X})}. \tag{24}$$

Lemma 3. $[\mathsf{X}_a, \mathsf{T}[g]] = \mathsf{T}[\Delta_a g]$, *hence* $\mathrm{ad}_{\mathsf{X}}^{\beta} \mathsf{T}[g] = \mathsf{T}[\boldsymbol{\Delta}^\beta g]$. *(Checked to* $6 \cdot 10^{-14}$*.)*

Lemma 4 (midpoint)**.** *For* $\Phi = \frac{1}{2}\boldsymbol{r}^\top M \boldsymbol{r} + \boldsymbol{b} \cdot \boldsymbol{r} + c$, $M = M^\top$, $\bar{\boldsymbol{r}} = \frac{1}{2}(\boldsymbol{r} + \boldsymbol{r}')$, $\boldsymbol{\Delta} = \boldsymbol{r} - \boldsymbol{r}'$: $\Phi(\boldsymbol{r}) - \Phi(\boldsymbol{r}') = (M\bar{\boldsymbol{r}} + \boldsymbol{b}) \cdot \boldsymbol{\Delta}$ *exactly.*

Proposition 5 (exact Weyl symbol)**.** *For quadratic* Φ_j *(here* $M_j = \sum_q w_{jq} \nabla^2 m_q$*),*

$$\sigma_{\mathsf{N}}(\bar{\boldsymbol{r}}, \boldsymbol{k}) = \sum_j \delta\big(\boldsymbol{k} - \boldsymbol{k}_j - M_j \bar{\boldsymbol{r}} - \boldsymbol{b}_j\big), \qquad \mathsf{N} = \sum_j \delta(\Pi_j - \boldsymbol{k}_j), \quad \Pi_j = \mathsf{Q}_j \mathsf{P} \mathsf{Q}_j^{\mathsf{H}} = \mathsf{P} - M_j \mathsf{X} - \boldsymbol{b}_j, \tag{25}$$

with commuting components ($[\Pi_{j,a}, \Pi_{j,b}] = \mathrm{i}(M_{j,ba} - M_{j,ab}) = 0$*) and no higher-order corrections: conjugation by a quadratic phase is metaplectic, i.e. the shear* $(\boldsymbol{r}, \boldsymbol{k}) \mapsto (\boldsymbol{r}, \boldsymbol{k} - \nabla\Phi)$ *[7]. The effective trajectory at* $\bar{\boldsymbol{r}}$ *is* $\boldsymbol{k}_j + M_j \bar{\boldsymbol{r}} + \boldsymbol{b}_j$*.*

Proposition 6 (commutator and Δ-forms)**.** *With* $g^\alpha = \sum_j w_j^\alpha \mathrm{e}^{\mathrm{i}\boldsymbol{k}_j \cdot \boldsymbol{\Delta}}$*,*

$$\begin{aligned} \mathsf{N} &= \sum_\alpha \frac{\mathrm{i}^{|\alpha|}}{\alpha!} \mathrm{ad}_{m_1}^{\alpha_1} \cdots \mathrm{ad}_{m_4}^{\alpha_4} \mathsf{T}[g^\alpha], \\ \mathsf{N} = \sum_n \mathsf{K}_n, \qquad K_n(\boldsymbol{r}, \boldsymbol{r}') &= \frac{\mathrm{i}^n}{n!} \sum_j \big[(M_j \bar{\boldsymbol{r}} + \boldsymbol{b}_j) \cdot \boldsymbol{\Delta}\big]^n \mathrm{e}^{\mathrm{i}\boldsymbol{k}_j \cdot \boldsymbol{\Delta}} = \sum_{|\beta| \le n} \bar{\boldsymbol{r}}^\beta g_{n\beta}(\boldsymbol{\Delta}). \end{aligned} \tag{26}$$

The kernel $\bar{\boldsymbol{r}}^\beta g(\boldsymbol{\Delta})$ *is the Weyl-symmetrised product* $2^{-|\beta|} \sum_{\gamma \le \beta} \binom{\beta}{\gamma} \mathsf{X}^\gamma \mathsf{T}[g] \mathsf{X}^{\beta - \gamma}$*. Every truncation is exactly Hermitian. Order* n *costs* $\binom{n+3}{3}$ *FFT pairs (4, 10, 20, 35), and the remainder is* $\le \sum_j (2\Phi_{\max})^{n+1}/(n+1)!$*.*

Proofs: Hadamard $\mathrm{e}^{\mathrm{i}A} T \mathrm{e}^{-\mathrm{i}A} = \sum_n \frac{\mathrm{i}^n}{n!} \mathrm{ad}_A^n T$ with commuting $m_q(\mathsf{X})$ and the multinomial theorem; for the Δ-form expand in the pair phase of Lemma 4. (Order 8 reproduces the dense N to $5 \cdot 10^{-12}$.) First order: $\mathrm{i}\sum_q [m_q, \mathsf{T}[g^{e_q}]] = \frac{\mathrm{i}}{2} \sum_{q,a} \{\partial_a m_q(\mathsf{X}), \mathsf{T}[\Delta_a g^{e_q}]\}$ needs 7 kernels and 4 FFT pairs.

Segmented Toeplitz form with derivative multipliers. For one segment, $T_\ell = \mathsf{T}[\sum_n a_{\ell n} \mathrm{e}^{\mathrm{i}\boldsymbol{k}_n \cdot \boldsymbol{\Delta}}]$ with multiplier τ_ℓ, and a quadratic residual $C_n \approx c(\boldsymbol{k}_n)$:

$$\mathsf{N}_\ell \approx T_\ell + \mathrm{i}\sum_{pq}\{\mathsf{X}_p, [\mathsf{X}_q, S_{\ell,pq}]\} - \tfrac{1}{2}\sum_{pqst}\{\mathsf{X}_p, \{\mathsf{X}_s, [\mathsf{X}_q, [\mathsf{X}_t, S_{\ell,pq,st}]]\}\}, \qquad \hat{S}_{\ell,pq} = c_{pq}\tau_\ell,\ \hat{S}_{\ell,pq,st} = c_{pq}c_{st}\tau_\ell, \tag{27}$$

$$\mathcal{F}\{\mathrm{i}\{\mathsf{X}_p, [\mathsf{X}_q, S]\}v\} = -\mathrm{i}\big[(\partial_{pq}\hat{S})V + 2(\partial_q \hat{S})(\partial_p V)\big]. \tag{28}$$

The transform count stays $2L$. However, τ_ℓ fills the whole $2N$ period and cannot be differenced, so it needs 35 stored derivative kernels per segment. $c(\boldsymbol{k})$ (degree 2 in $\boldsymbol{k}$) still leaves 0.066 rad. Not used for time-resolved sets.

Operator error vs. FFT pairs (8^3, 960 samples, rotated spirals; $\|\mathsf{N} - \mathsf{N}_{\text{approx}}\|_F / \|\mathsf{N} - \mathsf{N}^0\|_F$) [measured]:

method (FFT pairs)	$\Phi_{\max}$=0.3	1	3
Taylor of $\mathrm{e}^{-\mathrm{i}\Phi}$, P=3 (35)	$4.5 \cdot 10^{-4}$	$1.6 \cdot 10^{-2}$	0.37
Weyl n=1 (4)	$6.2 \cdot 10^{-2}$	0.21	0.61
Weyl n=2 (10)	$3.5 \cdot 10^{-3}$	$3.8 \cdot 10^{-2}$	0.34
Weyl n=4 (35)	$7.2 \cdot 10^{-6}$	$8.9 \cdot 10^{-4}$	$7.3 \cdot 10^{-2}$
SVD–Tucker L=12 (12)	$5.9 \cdot 10^{-4}$	$2.1 \cdot 10^{-3}$	$9.8 \cdot 10^{-3}$

$\|\mathsf{N} - \mathsf{N}^0\|/\|\mathsf{N}\| = 0.8, 2.8, 7.9\,\%$. No expansion has an FFT count independent of Φ: the Slepian number $\approx (\Phi_{\max}/\pi)c$ sets a floor of order L. Subdividing the samples (butterfly) increases the total basis count ($42 \to 51/80/132$ for 2/4/8 time bins at $\Phi \approx 9$).

1.8 Preconditioned plug-and-play iteration

$$x_{k+1} = \mathcal{D}\big(x_k - \alpha\, \mathsf{P}_k g_k\big), \qquad g_k = \nabla f(x_k), \qquad \mathsf{P}_k = \mathrm{ZMSHR1}\big(M;\, s_k = x_k - x_{k-1},\, u_k = g_k - g_{k-1}\big) \tag{29}$$

(P^{2}nP [8]): one operator application per iteration and a rank-one update of a base metric M. With spiral sampling N spans 10^4–10^6; the metric replaces density compensation. The base metric for the complex model is circulant, $\langle e_{\boldsymbol{k}}, \mathsf{N} e_{\boldsymbol{k}}\rangle = (\rho \circledast P_{\text{coil}})(\boldsymbol{k})$, with normalised maps ($\sum_c |S_c|^2 = 1$) and a floor on unsampled corners: $\lambda_{\max}(M\mathsf{N})$ goes from 38 to 9. The velocity model uses the pointwise Gauss–Newton blocks of Section 1.5.

Lagged correction. $\tilde{g}_k = \mathsf{N}_{\mathrm{s}}(x_k - x_a) + g(x_a)$, refreshing $x_a \leftarrow x_k$ when $\|x_k - x_a\| > \tau\|x_k\|$. This removes the surrogate bias at the fixed point. Cheap-only gradients are unsafe under a quasi-Newton metric: secant pairs from a mismatched operator act where N has small eigenvalues. 16^3 toy, $\Phi_{\max} = 3$, 100 iterations, identity denoiser; cost in plain-Toeplitz units, exact = 39 [measured]:

gradient	NRMSE	exact applications	cost
exact every iteration	0.213	100	15600
plain surrogate only	1.29	0	400
Weyl order 2 only	$7.7 \cdot 10^4$	0	12800
lagged plain, $\tau = 0.1$ / 0.05	0.218 / 0.214	12 / 19	2320 / 3440

This is 4.5–6.7× cheaper at equal error on the toy. With the reduced-segment surrogate on the real operator the gain is smaller and needs a mismatch $\lesssim 3\,\%$ (Section 1.6). It is not yet tested with the real denoiser.

1.9 Validation and limits

Truth data come from a GPU direct sum over a fine point cloud with exact (M_0, M_1, M_2), susceptibility B0, GNL and the full concomitant phase. It shares nothing with the operators and agrees with an f64 CPU sum to $1.2 \cdot 10^{-5}$ [measured]:

experiment	result
time-averaged LS per encode (R4, 60 s): no physics / B0 / B0+conc.	0.211 / 0.249 / 0.131 m/s
CFD case (RACLETTE n001), B0 + conc. + GNL, model-exact data	image $4.3 \cdot 10^{-3}$, velocity 2.4 mm/s
velocity model (Section 1.5) vs. complex + LLR	0.138 vs. 0.183 m/s

Limits.

- Concomitant fields are first order in G/B_0, computed from nominal or GIRF-filtered waveforms; eddy-current residuals have not been measured.
- Taylor stencils hold for $|\chi_j| \lesssim 0.7$ rad. At 0.55 T they need least-squares stencils inside the segments, which are not built.
- L grows with $\Delta f\, T_{\mathrm{ro}}$; long readouts pay directly in FFTs.
- Readout motion and intravoxel dephasing are not modelled.
- Operator error on typical images is not the relevant accuracy for a quasi-Newton iteration (Section 1.8).

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